Statistical Analysis Report: Maximum Heart Rate and Heart Disease Status

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GitHub Repository: https://github.com/sandipdey2/statistical-analysis-project

Overview

This report presents a statistical analysis of the Heart Failure Prediction dataset to examine whether maximum heart rate differs between patients with and without heart disease. The dataset is the combined Heart Failure Prediction Dataset curated by fedesoriano from multiple UCI sources (https://www.kaggle.com/datasets/fedesoriano/heart-failure-prediction).
The analysis focuses on descriptive patterns and a formal hypothesis test that can later support machine-learning or clinical-risk work. The complete project is available at https://github.com/sandipdey2/statistical-analysis-project.

Dataset Description

The dataset contains clinical records for 918 patients and 12 variables. Key variables examined include Age, Sex, ChestPainType, RestingBP, Cholesterol, MaxHR (maximum heart rate achieved), ExerciseAngina, and the binary target HeartDisease (0 = no disease, 1 = disease present). The data originate from several historical UCI heart-disease collections that were harmonized on common features. Numeric variables such as Age, RestingBP, and MaxHR are continuous; Sex, ChestPainType, and HeartDisease are categorical. Some Cholesterol values are recorded as zero and are treated as likely missing for interpretation purposes. The sample is large enough for reliable descriptive statistics and a two-sample test.

Methods

Descriptive statistics (means, standard deviations, and frequency counts) were computed for all key variables to characterize the sample before

formal testing. Three visual models were created: a histogram of MaxHR to show its overall distribution, a boxplot of MaxHR stratified by HeartDisease status, and a grouped bar chart of disease counts by Sex. The boxplot was judged most relevant to the analytical question. A Welch's two-sample t-test was selected because MaxHR is continuous, the two disease groups are independent, and the large sample sizes support the central-limit approximation. The null hypothesis stated that mean MaxHR is equal in both groups; the alternative stated that the means differ. Assumptions of independence and approximate normality of the sampling distribution were considered reasonable given the sample size (Huebner et al., 2024).

Results

Patients without heart disease (n = 410) had a mean maximum heart rate of 148.15 bpm (SD = 23.29). Patients with heart disease (n = 508) had a lower mean of 127.66 bpm (SD = 23.39). The Welch two-sample t-test produced a t-statistic of 13.23 and a p-value of approximately 1.43×10^{-36}, indicating that the observed difference is extremely unlikely under the null hypothesis of equal means. Figure 1 shows a unimodal distribution of MaxHR. Figure 2 (boxplot) visually confirms the downward shift in MaxHR among the disease group. Figure 3 shows a higher absolute number of disease cases among males. These patterns are consistent across the descriptive statistics and the formal test.

Interpretation for a Non-Technical Audience

People who have heart disease in this dataset tend to reach a lower maximum heart rate during exercise testing than people who do not have heart disease. On average, the difference is about 20 beats per minute. This does not mean that a low heart rate causes heart disease; it simply means the two groups differ on this measurement. The finding matches what doctors often observe clinically: patients with heart problems frequently have reduced exercise capacity. The results are useful for understanding the data but should not be used alone for medical decisions.

Limitations and Potential Bias

One clear limitation is the presence of zero values in the Cholesterol column, which almost certainly represent missing data rather than true

zero measurements; leaving them unchanged can slightly distort correlations that involve cholesterol. Another potential source of bias is the historical and multi-center origin of the data: recording practices, patient selection, and diagnostic criteria may have differed across the original studies, so the combined sample may not perfectly represent any single modern population. Both issues were noted but not fully corrected in this exploratory analysis.

References

Huebner, M., Vach, W., & le Cessie, S. (2024). Initial data analysis for longitudinal studies to build a solid foundation for reproducible analysis. *PLOS ONE, 19*(1), e0295726. https://doi.org/10.1371/journal.pone.0295726

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